Сетевые технологии

Адресация IPv4 и IPv6. Двойной стек

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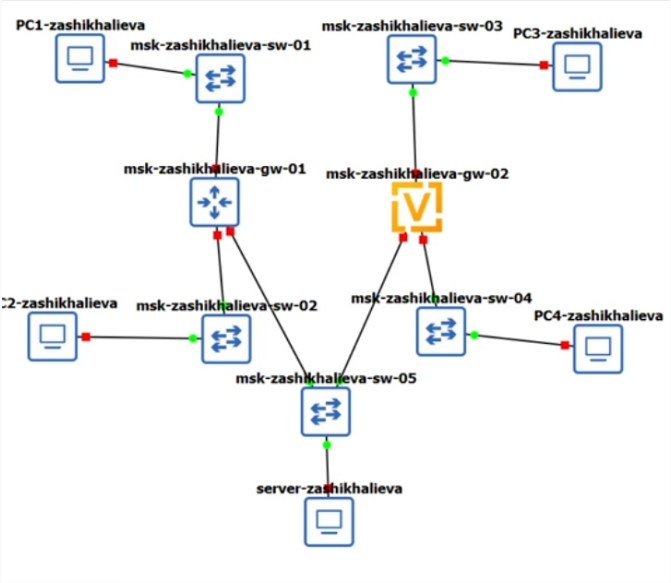
Цель работы

Основная цель

Изучить методы распределения IPv4/IPv6-адресов, разбиение сетей на подсети и настройку двойного стека в виртуальной лабораторной среде.

Настройка двойного стека адресации IPv4 и IPv6

Топология стенда



Конфигурация PC1

IPv4: 172.16.20.x

IPv6: автоматическое назначение SLAAC

Проверка командой `show ip`

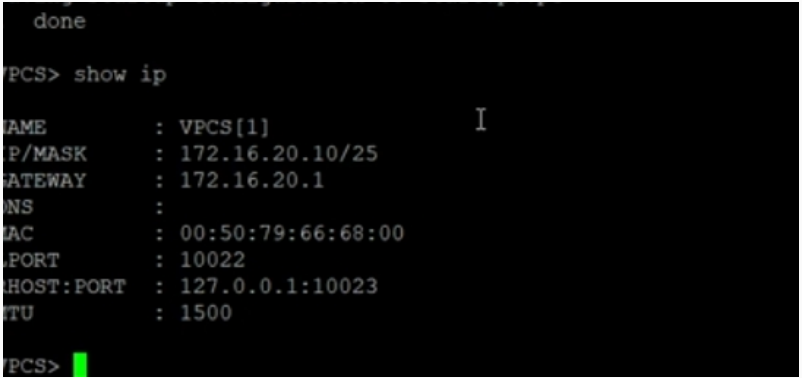


Рис. 2: PC1

Конфигурация PC2

IPv4: из диапазона 172.16.20.128/26

IPv6: SLAAC

```
done

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 172.16.20.10/25
GATEWAY     : 172.16.20.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 10022
RHOST:PORT  : 127.0.0.1:10023
MTU         : 1500

VPCS>
```

Рис. 3: PC2

Интерфейсы FRR

- eth0 — 172.16.20.1/25
- eth1 — 172.16.20.129/25
- eth2 — 64.100.1.1/24

```
[OK]
msk-zashikhalieva-gw-01#
msk-zashikhalieva-gw-01# configure terminal
msk-zashikhalieva-gw-01(config)# interface eth0
msk-zashikhalieva-gw-01(config-if)# ip address 172.16.20.1/25
msk-zashikhalieva-gw-01(config-if)# no shutdown
msk-zashikhalieva-gw-01(config-if)# exit
msk-zashikhalieva-gw-01(config)#
msk-zashikhalieva-gw-01(config)# interface eth1
msk-zashikhalieva-gw-01(config-if)# ip address 172.16.20.129/25
msk-zashikhalieva-gw-01(config-if)# no shutdown
msk-zashikhalieva-gw-01(config-if)# exit
msk-zashikhalieva-gw-01(config)#
msk-zashikhalieva-gw-01(config)# interface eth2
msk-zashikhalieva-gw-01(config-if)# ip address 64.100.1.1/24
msk-zashikhalieva-gw-01(config-if)# no shutdown
msk-zashikhalieva-gw-01(config-if)# exit
msk-zashikhalieva-gw-01(config)# exit
msk-zashikhalieva-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration
```

Ping и трассировка

- PC1 → PC2
- PC1 → Server

```
4 bytes from 172.16.20.138 icmp_seq=3 ttl=63 time=5.567 ms
4 bytes from 172.16.20.138 icmp_seq=4 ttl=63 time=4.557 ms
4 bytes from 172.16.20.138 icmp_seq=5 ttl=63 time=4.121 ms

PC5> ping 64.100.1.10

4 bytes from 64.100.1.10 icmp_seq=1 ttl=63 time=6.660 ms
4 bytes from 64.100.1.10 icmp_seq=2 ttl=63 time=3.471 ms
4 bytes from 64.100.1.10 icmp_seq=3 ttl=63 time=2.189 ms
4 bytes from 64.100.1.10 icmp_seq=4 ttl=63 time=3.648 ms
4 bytes from 64.100.1.10 icmp_seq=5 ttl=63 time=2.718 ms

PC5> trace 172.16.20.138
race to 172.16.20.138, 8 hops max, press Ctrl+C to stop
1   172.16.20.1   1.450 ms  0.801 ms  0.881 ms
2   *172.16.20.138   1.770 ms (ICMP type:3, code:3, Destination port unreachabl
)

PC5> trace 64.100.1.10
race to 64.100.1.10, 8 hops max, press Ctrl+C to stop
1   172.16.20.1   1.228 ms  0.695 ms  0.644 ms
2   *64.100.1.10   0.799 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC3

Адрес: 2001:db8:c0de:12::/64

PC3-zashikhalieva - PuTTY

```
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 2001:db8:c0de:12::a/64
PC1 : 2001:db8:c0de:12::a/64

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ipv6

NAME               : VPCS[1]
LINK-LOCAL SCOPE   : fe80::250:79ff:fe66:6802/64
GLOBAL SCOPE       : 2001:db8:c0de:12::a/64
DNS                :
ROUTER LINK-LAYER  :
MAC                : 00:50:79:66:68:02
LPORT              : 10044
RHOST:PORT         : 127.0.0.1:10045
MTU:               : 1500

VPCS>
```

PC4

Адрес: 2001:db8:c0de:13::/64

```
VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ipv6

NAME                  : VPCS[1]
LINK-LOCAL SCOPE      : fe80::250:79ff:fe66:6803/64
GLOBAL SCOPE          : 2001:db8:c0de:13::a/64
DNS                   :
ROUTER LINK-LAYER     :
MAC                   : 00:50:79:66:68:03
LPORT                 : 10046
RHOST:PORT            : 127.0.0.1:10047
MTU:                  : 1500
```

Сервер

IPv6: 2001:db8:c0de:11::/64

Двойной стек: IPv4 + IPv6

```
VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 64.100.1.10/24
GATEWAY     : 64.100.1.1
DNS         :
MAC         : 00:50:79:66:68:04
LPORT       : 10026
RHOST:PORT  : 127.0.0.1:10027
MTU         : 1500

VPCS> show ipv6

NAME               : VPCS[1]
LINK-LOCAL SCOPE   : fe80::250:79ff:fe66:6804/64
GLOBAL SCOPE       : 2001:db8:c0de:1::a/64
DNS                :
ROUTER LINK-LAYER  :
MAC                : 00:50:79:66:68:04
LPORT              : 10026
```

Назначение адресов и RA

```
yOS is a free software distribution that includes multiple components,
ou can check individual component licenses under /usr/share/doc/*/copyright
yos@msk-zashikhalieva-gw-02:~$ configure
edit]
yos@msk-zashikhalieva-gw-02# set interfaces ethernet eth0 address 2001:db8:c0de
12::1/64
edit]
yos@msk-zashikhalieva-gw-02# set service router-advert interface eth0 prefix 20
1:db8:c0de:12::/64
edit]
yos@msk-zashikhalieva-gw-02# set interfaces ethernet eth1 address 2001:db8:c0de
13::1/64
edit]
yos@msk-zashikhalieva-gw-02# set service router-advert interface eth1 prefix 20
1:db8:c0de:13::/64
edit]
yos@msk-zashikhalieva-gw-02# set interfaces ethernet eth2 address 2001:db8:c0de
11::1/64
edit]
yos@msk-zashikhalieva-gw-02# set service router-advert interface eth2 prefix 20
1:db8:c0de:11::/64
edit]
yos@msk-zashikhalieva-gw-02#
```

Рис. 11: VyOS IPv6

Проверка интерфейсов

```
[edit]
vyos@msk-zashikhalieva-gw-02# commit
[edit]
vyos@msk-zashikhalieva-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-zashikhalieva-gw-02# show interfaces
 ethernet eth0 {
     address 2001:db8:c0de:12::1/64
     hw-id 0c:ca:90:40:00:00
 }
 ethernet eth1 {
     address 2001:db8:c0de:13::1/64
     hw-id 0c:ca:90:40:00:01
 }
 ethernet eth2 {
     address 2001:db8:c0de:11::1/64
     hw-id 0c:ca:90:40:00:02
 }
 loopback lo {
 }
[edit]
vyos@msk-zashikhalieva-gw-02#
```

Ping PC4 → PC3 и Server

PC3-zasnikhalleva - PuTTY

```
2001:db8:c0de:13::a icmp6_seq=4 ttl=62 time=2.974 ms
2001:db8:c0de:13::a icmp6_seq=5 ttl=62 time=1.318 ms

VPCS> trace 2001:db8:c0de:13::a

trace to 2001:db8:c0de:13::a, 64 hops max
 1 2001:db8:c0de:12::1   0.928 ms  0.565 ms  0.603 ms
 2 2001:db8:c0de:13::a   2.048 ms  0.968 ms  1.705 ms

VPCS> ping 2001:db8:c0de:11::a

2001:db8:c0de:11::a icmp6_seq=1 timeout
2001:db8:c0de:11::a icmp6_seq=2 timeout
2001:db8:c0de:11::a icmp6_seq=3 timeout
2001:db8:c0de:11::a icmp6_seq=4 timeout
2001:db8:c0de:11::a icmp6_seq=5 timeout

VPCS> trace 2001:db8:c0de:11::a

trace to 2001:db8:c0de:11::a, 64 hops max
 1 2001:db8:c0de:12::1   0.871 ms  0.461 ms  0.490 ms
 2   *  3.899 ms  *

VPCS>
```

Рис. 13: Ping PC4

ICMPv6-трафик

No.	Time	Source	Destination	Protocol	Length	Info
13	5.018426	0c:91:02:ec:00:02	Private_66:68:04	ARP	60	Who has 64.100.1.10? Tell 64.100.1.1
14	5.019658	Private_66:68:04	0c:91:02:ec:00:02	ARP	60	64.100.1.10 is at 00:50:79:66:68:04
15	22.647798	2001:db8:c0de:11::a	2001:db8:c0de:12::a	ICMPv6	118	Echo (ping) request id=0xec8b, seq=1, hop limit=64 (reply
16	22.650176	2001:db8:c0de:12::a	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) reply id=0xec8b, seq=1, hop limit=62 (request
17	23.652132	2001:db8:c0de:11::a	2001:db8:c0de:12::a	ICMPv6	118	Echo (ping) request id=0xec8b, seq=2, hop limit=64 (reply
18	23.654058	2001:db8:c0de:12::a	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) reply id=0xec8b, seq=2, hop limit=62 (request
19	24.655106	2001:db8:c0de:11::a	2001:db8:c0de:12::a	ICMPv6	118	Echo (ping) request id=0xec8b, seq=3, hop limit=64 (reply

```
> Frame 16: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0
> Ethernet II, Src: 0c:ca:90:40:00:02 (0c:ca:90:40:00:02), Dst: Private_66:68:04 (00:50:79:66:68:04)
v Internet Protocol Version 6, Src: 2001:db8:c0de:12::a, Dst: 2001:db8:c0de:11::a
    0110 .... = Version: 6
  > .... 0000 0000 .... .... .... .... .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
    .... 0000 0000 0000 0000 0000 = Flow Label: 0x00000
    Payload Length: 64
    Next Header: ICMPv6 (58)
    Hop Limit: 62
  > Source Address: 2001:db8:c0de:12::a
  > Destination Address: 2001:db8:c0de:11::a
    [Stream index: 0]
v Internet Control Message Protocol v6
    Type: Echo (ping) reply (129)
    Code: 0
    Checksum: 0xbd7e [correct]
    [Checksum Status: Good]
    Identifier: 0xec8b
    Sequence: 1
    [Response To: 15]
    [Response Time: 2,378 ms]
  > Data (56 bytes)
```

Самостоятельное задание

Подсеть 1

- IPv4: 10.10.1.96/27
- IPv6: 2001:db8:1:1::/64

Подсеть 2

- IPv4: 10.10.1.16/28
- IPv6: 2001:db8:1:4::/64

Таблица адресов

Устройство	Интерфейс	IPv4	Маска	IPv6	Префикс
gw	eth0	10.10.1.97	/27	2001:db8:1:1::1	/64
gw	eth1	10.10.1.17	/28	2001:db8:1:4::1	/64
PC1	vpcs	10.10.1.100	/27	2001:db8:1:1::a	/64
PC2	vpcs	10.10.1.20	/28	2001:db8:1:4::a	/64

PC1

```
VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 10.10.1.100/27
GATEWAY     : 10.10.1.97
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 10008
RHOST:PORT  : 127.0.0.1:10009
MTU         : 1500

VPCS> show
```

Рис. 15: PC1 show ip

PC2

```
VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 10.10.1.100/27
GATEWAY     : 10.10.1.97
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 10008
RHOST:PORT  : 127.0.0.1:10009
MTU         : 1500
```

Рис. 16: PC2 show ip

Топология

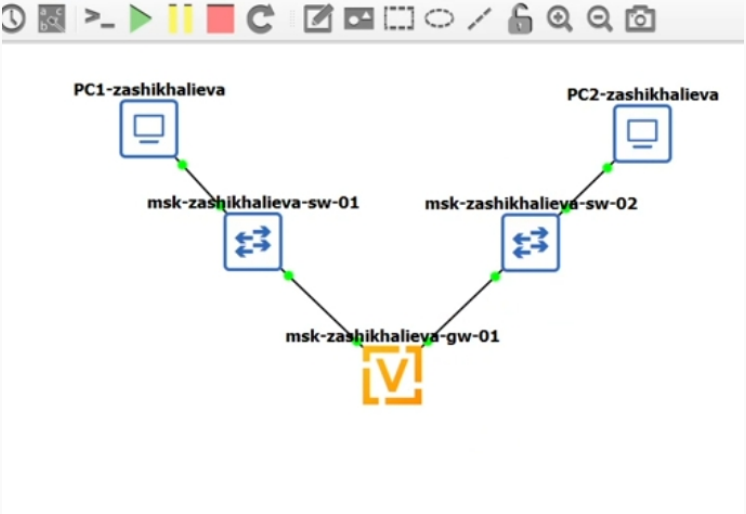


Рис. 17: Топология

Настройка маршрутизатора

```
VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 10.10.1.20/28
GATEWAY     : 10.10.1.17
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 10010
RHOST:PORT  : 127.0.0.1:10011
MTU         : 1500

VPCS> show ipv6

NAME               : VPCS[1]
LINK-LOCAL SCOPE   : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE       : 2001:db8:1:4::a/64
DNS                :
ROUTER LINK-LAYER  :
MAC                : 00:50:79:66:68:01
LPORT              : 10010
RHOST:PORT         : 127.0.0.1:10011
MTU:               : 1500
```

Рис. 18: VyOS interfaces

Проверки

```
vyos@msk-zashikhalieva-gw-01:~$ configure
[edit]
vyos@msk-zashikhalieva-gw-01# set interfaces ethernet eth0 address 10.10.1.97/27
[edit]
vyos@msk-zashikhalieva-gw-01# set interfaces ethernet eth0 address 2001:db8:1:1:
:1/64
[edit]
vyos@msk-zashikhalieva-gw-01# set service router-advert interface eth0 prefix 20
01:db8:1:1::/64
[edit]
vyos@msk-zashikhalieva-gw-01#
[edit]
vyos@msk-zashikhalieva-gw-01# set interfaces ethernet eth1 address 10.10.1.17/28
[edit]
vyos@msk-zashikhalieva-gw-01# set interfaces ethernet eth1 address 2001:db8:1:4:
:1/64
[edit]
vyos@msk-zashikhalieva-gw-01# set service router-advert interface eth1 prefix 20
01:db8:1:4::/64
[edit]
vyos@msk-zashikhalieva-gw-01# commit
```

Рис. 19: Ping PC1

Проверки

```
VPCS> ping 10.10.1.20

84 bytes from 10.10.1.20 icmp_seq=1 ttl=63 time=5.907 ms
84 bytes from 10.10.1.20 icmp_seq=2 ttl=63 time=3.593 ms
84 bytes from 10.10.1.20 icmp_seq=3 ttl=63 time=6.520 ms
84 bytes from 10.10.1.20 icmp_seq=4 ttl=63 time=3.712 ms
84 bytes from 10.10.1.20 icmp_seq=5 ttl=63 time=4.757 ms

VPCS> ping 2001:db8:1:4::a

2001:db8:1:4::a icmp6_seq=1 ttl=62 time=6.148 ms
2001:db8:1:4::a icmp6_seq=2 ttl=62 time=3.659 ms
2001:db8:1:4::a icmp6_seq=3 ttl=62 time=4.095 ms
2001:db8:1:4::a icmp6_seq=4 ttl=62 time=3.025 ms
```

Рис. 20: IPv6 ICMP

Итоги работы

Основные результаты

- Выполнено детальное разбиение IPv4/IPv6-сетей
- Настроены двойные стеки на ПК и маршрутизаторах
- Проверена маршрутизация, ARP/ND, ICMPv4/ICMPv6
- Подтверждена корректная работа сетей и взаимодействие подсетей только через маршрутизатор